

Project Proposal: Prediction of Potential Wildfire Regions Using Heat Map Analysis and Image Recognition

Alok Pratap Singh

Department of Analytics, Georgia Tech University

June 29, 2025

1 Introduction

Wildfires represent an increasingly severe global crisis, causing extensive damage to ecosystems, economies, and human lives. The destruction of vast forest and grassland areas, displacement of communities, pervasive air pollution, and acceleration of climate change underscore the urgent need for more effective wildfire management strategies. A critical component of such strategies is the ability to accurately and promptly detect and predict wildfire ignition. Current approaches, which often rely on infrequent satellite imagery, ground patrols, and human observation, are frequently hampered by limitations in real-time capability, spatial coverage, and inherent dangers, especially in remote and inaccessible terrains.

This project proposal outlines a comprehensive vision for a full-fledged wildfire prediction system using heat map analysis and image recognition, incorporating drone-based deployment for analysis over large and hard-to-access areas. It is important to note, however, that the scope of the immediate project implementation will focus on a primitive analysis using readily available, simpler datasets to validate core methodologies and demonstrate proof-of-concept before scaling to the full complexity of real-time drone data integration. This initial phase will lay the groundwork and validate the theoretical approach for the more advanced system described herein.

2 Problem Statement

The central problem this project addresses is the inadequate capacity for early and autonomous wildfire prediction in vast, difficult-to-access natural landscapes. Existing wildfire monitoring systems are predominantly reactive, designed to detect fires once they have already ignited, rather than predicting where and when they are likely to start. This inherent lag between ignition and detection often allows nascent fires to grow into uncontrollable conflagrations, leading to catastrophic outcomes. The specific limitations that necessitate a novel solution include:

- **Inaccessibility of Remote Terrains:** Many regions most susceptible to wildfires are geographically challenging, comprising dense forests, rugged mountains, or remote grasslands. These areas are extremely difficult for human personnel or ground-based equipment to monitor effectively, safely, or quickly. This inaccessibility directly translates to delayed detection and response times.

- **Limited Scalability of Current Methods:** Conventional monitoring techniques, such as fixed sensor networks or manual patrols, cannot realistically scale to cover the immense geographical extents of fire-prone areas. This results in significant surveillance gaps, allowing fires to emerge unnoticed in vast, unmonitored zones.
- **Lag in Satellite-Based Monitoring:** While satellite imagery provides broad coverage, its utility for real-time early detection is constrained by its refresh rate and resolution limitations. Crucial early indicators, such as localized temperature spikes or subtle changes in vegetation health, can often be missed between satellite passes.
- **Complexity of Risk Factor Interpretation:** Identifying the nuanced environmental cues that precede a wildfire (e.g., specific moisture levels in vegetation, micro-climate temperature anomalies, or the presence of specific dry fuel types) requires sophisticated analytical capabilities. Human visual inspection over large areas is prone to oversight and cannot consistently process the multivariate data necessary for accurate prediction.
- **High Resource Intensity:** The continuous monitoring of large areas using traditional methods demands substantial financial investment in infrastructure, maintenance, and personnel, making it unsustainable for widespread, long-term implementation.

Our project directly confronts these challenges by proposing an autonomous, drone-based predictive system. By leveraging machine learning to process high-resolution thermal and visible light imagery, we aim to transform wildfire management from a reactive response to a proactive, predictive approach. This constitutes a problem in predictive analytics and spatial classification, analogous to predicting complex environmental phenomena or identifying high-risk zones in other domains, but specifically tailored to the dynamic and heterogeneous nature of wildfire initiation.

3 Data Collection

The successful implementation of this project hinges upon the strategic acquisition and integration of diverse datasets, primarily focusing on information obtainable via drone deployment. For the immediate project (primitive analysis phase), publicly available, simpler datasets will be utilized to validate core model ideas and data processing pipelines. The following subsections detail the ideal data for a full-fledged system and specify the datasets that will be used for the current project's analysis.

3.1 Thermal Imagery Data

This will serve as the cornerstone for heat map analysis in the full system. Drones equipped with high-resolution thermal infrared (IR) cameras would capture the emitted radiation, allowing for the precise detection of temperature anomalies on the ground.

- **Ideal Data for Full System:** High-resolution, georeferenced thermal imagery acquired directly from drone flights over diverse wildland terrains, capturing temporal changes in temperature.
- **Data for This Project:** For preliminary model development, we will primarily utilize:
 - *FIRMS (Fire Information for Resource Management System)*: Specifically, the near real-time active fire/thermal hotspot data (from MODIS and VIIRS sensors) [1], [2].

While satellite-based and coarser resolution than drone data, it provides crucial thermal anomaly information for historical events that can be used for initial analysis of hot spot detection.

- *FLAME 3 Dataset*: This dataset provides radiometric thermal UAV imagery focused on wildfire management, including raw thermal images and georeferenced overhead thermal imagery, captured by UAVs during prescribed burns [3]. This offers valuable, albeit localized, drone-like thermal data for understanding thermal signatures of fires and their progression.

3.2 Visible Light Imagery (RGB)

Complementing thermal data, standard high-resolution RGB cameras mounted on drones would capture visual information detailing vegetation type, density, health, and terrain characteristics for the full system.

- **Ideal Data for Full System:** Continuous, high-resolution RGB imagery from drone patrols, enabling fine-grained analysis of vegetation stress, fuel load, and atmospheric conditions (e.g., smoke presence).
- **Data for This Project:** For initial training and testing of image recognition components, we will utilize:
 - *The Wildfire Dataset (Kaggle)*: This large collection of aerial and ground-based RGB images (with 'wildfire' and 'no wildfire' classes) is curated from diverse online platforms and is suitable for deep learning-based forest fire detection [4]. It provides a good starting point for training image classification models on visual fire indicators.
 - *Digital Aerial Imagery from 2010-SOUTH TRAIL wildfire boundary (RCDS Data Repository)*: This dataset includes raw RGB and multispectral imagery collected by UAVs over a wildfire boundary [5]. It offers higher resolution and more relevant visual data for vegetation and fuel analysis, although from specific fire events rather than broad area monitoring.

3.3 Geospatial Data (GIS)

Pre-existing geospatial datasets would be integrated to provide static environmental context for the dynamic drone data in a full system.

- **Ideal Data for Full System:** Comprehensive, up-to-date GIS layers covering topography, detailed land cover (e.g., fuel models), historical fire perimeters, and infrastructure, all at a high spatial resolution.
- **Data for This Project:** For the primitive analysis, readily available and widely used geospatial data will be incorporated:
 - *LANDFIRE Program data from USGS*: This provides national geospatial layers on vegetation, fuel, and disturbance, which are essential for understanding land cover and fuel types [6].
 - *MTBS (Monitoring Trends in Burn Severity) and NIFC data*: These sources provide historical fire perimeters and incident locations, crucial for defining ground truth for training and evaluation [6], [7].

- *OpenStreetMap Data*: For basic infrastructure mapping (roads, settlements) in selected study areas.

3.4 Weather Data

Meteorological parameters are pivotal contextual features for a complete wildfire prediction system, ideally incorporating real-time, hyperlocal data.

- **Ideal Data for Full System:** Real-time, fine-resolution weather data (temperature, humidity, wind speed/direction, precipitation, lightning strikes) integrated from on-board drone sensors and dense ground weather stations.
- **Data for This Project:** For the primitive analysis, we will focus on historical weather patterns and fire weather indices:
 - *Kaggle Forest Fires Data Set*: This dataset includes meteorological data (temperature, relative humidity, wind, and rain) alongside burned area, providing a direct link between simple weather parameters and fire occurrence for a specific region [8].
 - *OpenWeather Fire Weather Index API*: For access to the Canadian Forest Service Fire Weather Index (FWI) data, which is a widely used composite index for fire danger [9]. This offers a simplified yet effective proxy for fire-conducive weather conditions.

3.5 Historical Wildfire Incident Data

Detailed records of past wildfire events are invaluable for training and validating predictive models.

- **Ideal Data for Full System:** Comprehensive, granular records of past wildfires, including precise ignition points, fire spread polygons, and detailed environmental conditions at the time of ignition and throughout the fire’s duration.
- **Data for This Project:** We will leverage existing national and regional databases:
 - *National Interagency Fire Center (NIFC) and USGS Wildland Fire Science data*: These sources provide aggregated statistics, incident reports, and maps of historical wildfire occurrences, which will be critical for defining the target variable (wildfire ignition areas) and for overall model validation [6], [7].

The dataset size for this primitive analysis will be constrained by the availability and accessibility of these selected public datasets. Our emphasis will remain on acquiring high-quality, representative data that allows for the fundamental testing of our analytical approaches and feature engineering pipelines. The novelty of our approach lies in the conceptual framework and the synergistic integration of these diverse data types, even if initially at a coarser resolution, to build a predictive model.

4 Data Analysis and Model Fitting (Methodology)

Our methodology involves a structured, multi-stage data analysis and model fitting process, drawing upon established techniques from data analysis and machine learning to build a robust predictive system.

4.1 Data Preprocessing and Feature Engineering

Before model training, raw data will undergo extensive preprocessing and feature engineering:

- **Thermal Imagery Processing:** Raw thermal images will be rectified, georeferenced, and converted into quantitative temperature maps. Advanced image processing techniques, such as spatial smoothing (e.g., Gaussian filters) and outlier detection, will be applied to minimize sensor noise and environmental interference. Key features extracted from thermal data will include:
 - *Average Temperature:* Mean temperature within defined grid cells.
 - *Maximum Temperature:* Peak temperature within a grid cell, indicative of potential hotspots.
 - *Temperature Variance:* Spatial variability of temperature within a cell, suggesting non-uniform heating.
 - *Rate of Temperature Change:* Temporal gradient of temperature for persistent monitoring, indicating a heating trend.
- **Visible Light Imagery Processing and Image Recognition:** Standard RGB images will be processed using Convolutional Neural Networks (CNNs) for object detection and classification relevant to fire risk:
 - *Vegetation Health Indices:* Calculation of spectral indices like Normalized Difference Vegetation Index (NDVI) or Enhanced Vegetation Index (EVI) to quantify vegetation greenness and moisture content, with lower values indicating drier, more flammable vegetation.
 - *Fuel Load Estimation:* Classification of vegetation types and estimation of biomass density (e.g., identifying areas with high concentrations of dry grass, deadfall, or dense shrubbery) using semantic segmentation or object detection models.
 - *Atmospheric Conditions and Early Smoke Detection:* While primary focus is on pre-fire, CNNs can also be trained to detect subtle smoke plumes as an immediate alert mechanism, or to identify visibility issues.
- **Geospatial and Weather Data Integration:** All processed drone-derived features (thermal and image recognition outputs) will be spatially joined with the static GIS data and dynamic weather parameters. Features derived from GIS will include slope, aspect, elevation, proximity to human activity/infrastructure, and historical fire frequency. Weather data will be integrated as time-series features (e.g., current temperature, cumulative rainfall over past 24/48 hours, wind vector components).
- **Target Variable Definition:** The problem will be framed as a binary classification task. Each defined grid cell (e.g., $10\text{m} \times 10\text{m}$ or $25\text{m} \times 25\text{m}$) within the monitored area will be assigned a label: '1' if it is a "potential wildfire ignition region" (based on historical ignition points or expert-identified high-risk zones) and '0' for "safe region."

4.2 Model Selection and Fitting

We will explore and compare a range of machine learning models, chosen for their suitability in handling complex, heterogeneous datasets and their interpretability for real-world application:

- **Traditional Supervised Classification Models:**

- *Logistic Regression*: To serve as a robust baseline, providing insights into the linear separability of features related to wildfire risk.
- *Support Vector Machines (SVMs)*: Particularly effective for high-dimensional feature spaces, SVMs will be used to identify optimal hyperplanes that separate fire-prone from non-fire-prone regions.
- *Ensemble Methods (Random Forests, Gradient Boosting Machines - GBM)*: These powerful non-linear models are highly robust to noisy data and capable of capturing complex interactions between features. They will be critical for identifying the most important contributing factors to wildfire risk through feature importance analysis.

- **Deep Learning Models:**

- *Convolutional Neural Networks (CNNs)*: For the direct analysis of raw thermal and RGB images. These models will learn hierarchical features directly from the pixel data, identifying subtle patterns indicative of fire risk that might be missed by manual feature engineering.
- *Multi-Modal Neural Networks*: An advanced approach where separate branches of a neural network handle different data modalities (e.g., one branch for image features, another for tabular geospatial/weather data), with their outputs concatenated and fed into a final classification layer. This allows for end-to-end learning from diverse data sources.

The output of the chosen model will be a probability score for each grid cell, indicating the likelihood of it becoming a wildfire ignition point. These probabilities will be spatially aggregated and visualized as a predictive heat map.

5 Evaluation Strategies

Rigorous evaluation is paramount to ascertain the model’s reliability, accuracy, and practical utility in a mission-critical application like wildfire prediction. Our evaluation strategy will encompass both standard classification metrics and specialized spatial accuracy assessments.

- **Cross-Validation**: To ensure the model’s generalizability and guard against overfitting, K-fold cross-validation will be extensively employed during the training phase. This involves partitioning the dataset into K subsets, training the model on K-1 subsets, and validating on the remaining one, iteratively.
- **Performance Metrics for Classification**:
 - *Accuracy*: Overall proportion of correctly classified instances.
 - *Precision*: The proportion of correctly predicted positive cases (wildfire risk) out of all predicted positive cases. High precision minimizes false alarms.
 - *Recall (Sensitivity)*: The proportion of correctly predicted positive cases out of all actual positive cases. High recall is critical to minimize missed wildfire opportunities (false negatives), which is a high-priority concern in this domain.

- *F1-score*: The harmonic mean of precision and recall, providing a balanced measure of the model’s performance, especially when class distribution is imbalanced.
 - *Receiver Operating Characteristic (ROC) Curve and Area Under the Curve (AUC)*: The ROC curve plots the true positive rate against the false positive rate at various threshold settings. AUC provides a single scalar value representing the model’s ability to discriminate between positive and negative classes, with a higher AUC indicating better performance.
 - *Confusion Matrix*: A detailed visualization of true positives, true negatives, false positives, and false negatives, providing insights into the types of errors made by the model. Minimizing false negatives will be prioritized, even if it entails a slight increase in false positives (which can be managed by further on-ground investigation).
- **Spatial Accuracy Assessment:** Beyond conventional metrics, the geographical precision of predictions is crucial:
 - *Geographical Distance Analysis*: We will measure the average and maximum geographical distances between predicted high-risk centroids and actual historical wildfire ignition points. A smaller average distance indicates higher spatial accuracy.
 - *Containment Rate*: Assessing what percentage of actual wildfire ignition points fall within a predicted high-risk zone (e.g., within the top 10% of predicted probability areas).
 - *Spatial Overlap Analysis*: Using GIS tools to quantify the degree of overlap between predicted high-risk areas and actual historical fire perimeters.
 - **Model Interpretability and Feature Importance:** For models like Random Forests or Gradient Boosting Machines, we will analyze feature importance scores. This will help in understanding which specific environmental factors (e.g., elevated temperatures, low vegetation moisture, high wind speeds, specific terrain features) are most predictive of wildfire ignition. This interpretability not only validates the model’s logic but also provides actionable insights for wildfire prevention strategies.
 - **Robustness Testing:** The model will be tested against novel datasets representing varying geographical regions, seasons, and weather conditions not seen during training, to assess its robustness and generalizability across diverse environments.

We anticipate that the rigorously evaluated model will demonstrate strong predictive capabilities, particularly in identifying areas with high potential for wildfire ignition. The interpretability of the model’s decisions will enhance trust and facilitate targeted preventative actions.

6 Feasibility and Real-World Application

The proposed project is highly feasible, benefiting from rapid advancements in several convergent technologies. Drone capabilities, including endurance, payload capacity for thermal and RGB sensors, and autonomous flight planning, have matured significantly. Similarly, the computational power required for on-board processing (or rapid cloud-based analysis post-collection) and the sophistication of machine learning algorithms (especially deep learning for image analysis) are well within current technological reach. Data accessibility from various public and scientific repositories also supports the data collection phase.

The real-world implications and potential for positive impact are substantial:

- **Proactive Fire Management:** The primary benefit is the shift from reactive firefighting to proactive prevention. By identifying potential hotspots days or even hours in advance, authorities can initiate controlled burns in a safe and managed manner, significantly reducing the fuel load and mitigating the risk of large-scale uncontrolled wildfires.
- **Optimized Resource Deployment:** Firefighting resources (personnel, equipment, air support) are expensive and limited. Predictive heat maps allow for the strategic pre-positioning of these resources in anticipated high-risk zones, drastically improving response times and efficiency.
- **Enhanced Public Safety and Early Warning:** Timely and accurate predictions enable the issuance of early warnings and facilitate pre-emptive evacuation orders for communities located in or near predicted high-risk areas, thereby saving lives and property.
- **Targeted Surveillance:** The model’s output can guide human patrols and additional ground-based sensor deployments to specific, high-priority areas, optimizing surveillance efforts and reducing human exposure to hazardous environments.
- **Environmental Preservation:** By preventing or minimizing the scale of wildfires, the project contributes significantly to the preservation of biodiversity, forest ecosystems, and reduction of carbon emissions associated with large fires.

The envisioned deployment model, where the predictive model is loaded onto drones for autonomous, on-site analysis, ensures rapid data processing and immediate actionable intelligence for remote environments. The scheduled reports or real-time alerts generated by the system will directly inform and enhance existing wildfire management protocols, leading to a more efficient, safer, and ultimately more effective approach to mitigating the devastating impact of wildfires.

References

- [1] NASA Earthdata. “FIRMS — NASA Earthdata.” Accessed on June 29, 2025. (), [Online]. Available: <https://www.earthdata.nasa.gov/data/tools/firms> (visited on 06/30/2025).
- [2] NASA Earthdata LANCE. “LANCE — FIRMS - Active Fire Data.” Accessed on June 29, 2025. (), [Online]. Available: https://firms.modaps.eosdis.nasa.gov/active_fire/ (visited on 06/30/2025).
- [3] B. Hopkins, L. O’Neill, M. Marinaccio, *et al.*, “FLAME 3 Dataset: Unleashing the Power of Radiometric Thermal UAV Imagery for Wildfire Management,” *Preprint*, 2024, Accessed on June 29, 2025. [Online]. Available: <https://par.nsf.gov/biblio/10588422-flame-radiometric-thermal-uav-imagery-wildfire-management>.
- [4] I. El-Madafri, M. Peña, and N. Olmedo-Torre, “The Wildfire Dataset: Enhancing Deep Learning-Based Forest Fire Detection with a Diverse Evolving Open-Source Dataset Focused on Data Representativeness and a Novel Multi-Task Learning Approach,” *Forests*, vol. 14, no. 9, p. 1697, 2023, Accessed on June 29, 2025. DOI: 10.3390/f14091697. [Online]. Available: <https://www.kaggle.com/datasets/elmadafri/the-wildfire-dataset>.
- [5] V. Marie, A. Zaiats, A. Roser, *et al.* “Digital aerial imagery (RGB and multispectral) from the 2010-SOUTH TRAIL wildfire boundary near Hammett Idaho USA-2022.” Accessed on June 29, 2025. (2023), [Online]. Available: <https://data.nkn.uidaho.edu/dataset/digital-aerial-imagery-rgb-and-multispectral-2010-south-trail-wildfire-boundary-near-hammett> (visited on 06/30/2025).

- [6] U.S. Geological Survey. “Wildland Fire Science — U.S. Geological Survey - USGS.gov.” Accessed on June 29, 2025. (), [Online]. Available: <https://www.usgs.gov/centers/eros/science/wildland-fire-science> (visited on 06/30/2025).
- [7] National Interagency Fire Center. “Fire Information — National Interagency Fire Center.” Accessed on June 29, 2025. (), [Online]. Available: <https://www.nifc.gov/fire-information> (visited on 06/30/2025).
- [8] P. Cortez and A. Morais. “Forest Fires Data Set - Kaggle.” Accessed on June 29, 2025. Original paper: Cortez, P., Morais, A. (2007). A data mining approach to predict forest fires using meteorological data. In J. Neves, M. F. Santos, J. Machado (Eds.), *New Trends in Artificial Intelligence* (pp. 512-523). Portuguese Conference on Artificial Intelligence (EPIA). (), [Online]. Available: <https://www.kaggle.com/datasets/elikplim/forest-fires-data-set> (visited on 06/30/2025).
- [9] OpenWeather. “Fire Weather Index API - OpenWeather.” Accessed on June 29, 2025. (), [Online]. Available: <https://openweathermap.org/api/fire-index-api> (visited on 06/30/2025).